

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – Technical Presentation

|                  |                                                                                                                                                          |               |                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------|
| Course Code:     | DSA0302                                                                                                                                                  | Course Title: | Natural Language Processing                |
| CO Assessed:     | CO1 – Imitation (L1)                                                                                                                                     | Assessment:   | Assessment Tool 2 – Technical Presentation |
| Weightage:       | 30% of CIA                                                                                                                                               |               |                                            |
| CO1 Statement:   | Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3) |               |                                            |
| Student Name:    | M.SIVA ANJANEYULU                                                                                                                                        | Reg. No.:     | 192424419                                  |
| Section / Batch: | AI&DS(2024-2028)                                                                                                                                         | Date:         | 15/07/2026                                 |

### Code of Conduct

I M. SIVA ANJANEYULU(192424419) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: m. siva anjaneyulu

### Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

| Section / Topic                                                                 | Focus                                           | Q Nos |
|---------------------------------------------------------------------------------|-------------------------------------------------|-------|
| Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | NLP Models, Language Processing Evolution       | TP-01 |
| Comparative Study of NLP Models and Algorithms                                  | Model Analysis, Algorithm Comparison            | TP-02 |
| Regular Expressions for Real-World Text Processing                              | Pattern Matching, Information Extraction        | TP-03 |
| Designing Efficient Regular Expression Patterns for Information Extraction      | Text Mining, Data Extraction Techniques         | TP-04 |
| Finite State Automata in Natural Language Processing                            | State Transitions, Language Recognition         | TP-05 |
| Building a Finite State Automaton for Token Recognition                         | Lexical Analysis, Token Validation              | TP-06 |
| Morphology and Word Formation in Natural Languages                              | Morpheme Analysis, Word Structure Processing    | TP-07 |
| Inflectional Morphology: Understanding Grammatical Variations                   | Grammatical Analysis, Word Normalization        | TP-08 |
| Derivational Morphology and Vocabulary Expansion                                | Word Generation, Semantic Transformation        | TP-09 |
| Comparative Analysis of Inflectional and Derivational Morphology                | Morphological Classification, Language Analysis | TP-10 |

|                                                                               |                                                |       |
|-------------------------------------------------------------------------------|------------------------------------------------|-------|
| Finite-State Morphological Parsing Techniques                                 | Morphological Parsing, Finite-State Processing | TP-11 |
| Morphological Analysis for Indian Languages                                   | Language-Specific Morphology, NLP Challenges   | TP-12 |
| Porter Stemmer: Design, Working Mechanism, and Applications                   | Stemming Algorithms, Text Preprocessing        | TP-13 |
| Stemming vs Morphological Parsing: A Comparative Study                        | Word Reduction, Morphological Processing       | TP-14 |
| Integrated NLP Pipeline: From Regular Expressions to Morphological Processing | End-to-End Language Processing Pipeline        | TP-15 |

## Annexure A – Technical Presentation

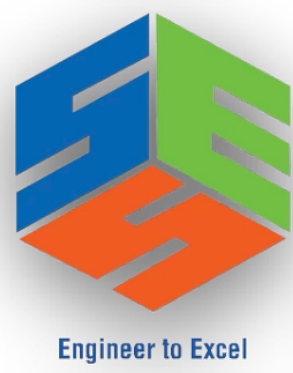
| Team No. | Technical Presentation Title                                                    | Description                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP | Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.              |
| 2        | Comparative Study of NLP Models and Algorithms                                  | Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.                             |
| 3        | Regular Expressions for Real-World Text Processing                              | Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.                 |
| 4        | Designing Efficient Regular Expression Patterns for Information Extraction      | Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.                    |
| 5        | Finite State Automata in Natural Language Processing                            | Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.                              |
| 6        | Building a Finite State Automaton for Token Recognition                         | Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.                                    |
| 7        | Morphology and Word Formation in Natural Languages                              | Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.                                             |
| 8        | Inflectional Morphology: Understanding Grammatical Variations                   | Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.                                               |
| 9        | Derivational Morphology and Vocabulary Expansion                                | Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.                                    |
| 10       | Comparative Analysis of Inflectional and Derivational Morphology                | Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.                                                   |
| 11       | Finite-State Morphological Parsing Techniques                                   | Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.                                   |
| 12       | Morphological Analysis for Indian Languages                                     | Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.        |
| 13       | Porter Stemmer: Design, Working Mechanism, and Applications                     | Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.                                      |
| 14       | Stemming vs Morphological Parsing: A Comparative Study                          | Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.                                         |
| 15       | Integrated NLP Pipeline: From Regular Expressions to Morphological Processing   | Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing. |

## Rubrics for Calculation

| Technical Presentation     |           |                                                                    |                                                                |                                                            |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15        | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |
| Professionalism            | 10        | Exemplary professionalism, leadership, and ethical awareness       | Professional conduct with effective team collaboration         | Basic professionalism; uneven team participation           | Lacks professionalism; minimal contribution or ethical awareness    |

| Team No. / Criteria Level | Technical Content Accuracy | Problem Identification | Use of Tools | Communication | Professionalism | Total % |
|---------------------------|----------------------------|------------------------|--------------|---------------|-----------------|---------|
|                           |                            |                        |              |               |                 |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |



**SIMATS ENGINEERING**  
**Saveetha Institute of Medical and Technical Sciences**  
**Chennai-602105.**

# **REGULAR EXPRESSIONS FOR REAL-WORLD TEXT PROCESSING**

DSA0302 - Natural Language Processing

**Supervisor**

**DR.T.DEVI,**

**SIMATS Engineering**

**Presented By**

**M.SIVA ANJANEYULU**

**REG NO :192424419**

# INTRODUCTION

- Regular Expressions (Regex) are sequences of characters used to define search patterns.
- They help search, match, extract, and replace text efficiently.
- Regex is supported by programming languages like Python, Java, JavaScript, and C#.
- It is widely used in text processing and data validation.
- Regex reduces manual effort and improves accuracy





# Data Validation Using Regular Expressions

- **Data validation** is the process of checking whether input data follows a required format.
- Regular Expressions (Regex) verify data before it is accepted or stored.
- They help prevent invalid, incomplete, or incorrectly formatted input.
- Regex ensures data consistency, accuracy, and reliability.

## Common Validation Examples

- **Email Address** – student@example.com
- **Phone Number** – 9876543210
- **Password Strength** – Minimum length with letters, numbers, and special characters
- **Student ID / Register Number** – Matches the required institutional format
- **Date Format** – DD/MM/YYYY or YYYY-MM-DD

# Real-World Applications of Regular Expressions

- Resume information extraction
- Email and phone number validation
- Password strength verification
- Product search using keywords
- Log file analysis
- Data cleaning and preprocessing
- Web scraping and text mining Form input validation



# Implementation with Examples

```
import re
```

```
email = "student123@gmail.com"
```

```
pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'
```

```
if re.match(pattern, email):
```

```
    print("Valid Email")
```

```
else:
```

```
    print("Invalid Email")
```

Output :

Valid Email

```
main.py  [Icons] [Share] [Run] [Output]
1  import re
2
3  text = """
4  Name: John
5  Email: john123@gmail.com
6  Phone: 9876543210
7  Date: 15/07/2026
8  Website: https://www.google.com
9  """
10
11 # Email Extraction
12 emails = re.findall(r'[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}', text)
13 print("Emails:", emails)
14
15 # Phone Number Extraction
16 phones = re.findall(r'\d{10}', text)
17 print("Phone Numbers:", phones)
18
19 # Date Extraction
20 dates = re.findall(r'\d{2}/\d{2}/\d{4}', text)
21 print("Dates:", dates)
22
23 # URL Extraction
24 urls = re.findall(r'https?://\S+', text)
25 print("URLs:", urls)
26
27 # Email Validation
```

Output

```
Emails: ['john123@gmail.com']
Phone Numbers: ['9876543210']
Dates: ['15/07/2026']
URLs: ['https://www.google.com']
Valid Email

=== Code Execution Successful ===
```

# Advantages and Limitations of Regular Expressions

## Advantages:

- Fast and efficient for searching and matching text patterns.
- Simplifies complex text-processing tasks.
- Validates user input such as emails, phone numbers, and passwords.

## Limitations:

- Complex regex patterns can be difficult to read and understand.
- Debugging long expressions can be challenging.
- Poorly designed regex may reduce performance on large datasets.

# Conclusion and Future Scope

## Conclusion

- Regular Expressions are powerful tools for text processing and pattern matching.
- They automate validation, extraction, and searching tasks.
- Regex improves efficiency, accuracy, and productivity in software applications.

## Future Scope

- AI-powered text processing.
- Natural Language Processing (NLP).
- Big Data analytics.
- Advanced cybersecurity applications.
- Intelligent document processing and automation.



